Warnings, Errors, and Getting Help

Recommended reading and watching

- https://stackoverflow.com/a/5963610/3187973
  - https://github.com/jennybc/debugging#video

Messages, warnings, and errors

- *Message*: Things are fine, but here's some information you should know
- *Warning*: Uhhhh I'm gonna keep going, but maybe this isn't what you want
- *Error*: Nope. I'm stopping here. You need to fix the thing.

Where to go for help

- Isolate the problem
- Restart your session: Session → Restart R
- Local help: `?fun`
- StackOverflow: https://stackoverflow.com/questions/tagged/r
  - CRAN → BugReports (usually GitHub Issues)

```r
library(lubridate)

some_data = c('June 2002', 'May 15, 2007', 'August 2007')
mdy(some_data)
parse_date_time(some_data, orders = 'mdY')
```

Let's try SO: https://stackoverflow.com/search?q=%5BR%5D+lubridate+month-year

```r
parse_date_time(some_data, orders = c('mY', 'mdY'))
```

Here's another one:

```r
more_data = c('May 7, 2017', 'May 19, 2017', 'May Fifth, 2017')
mdy(more_data)
```

SO doesn't seem so helpful: https://stackoverflow.com/search?q=%5BR%5D+lubridate+written+days

Let's check the CRAN page for `lubridate`: https://cran.r-project.org/web/packages/lubridate/index.html

Trying a couple of searches gives us a promising result: https://github.com/tidyverse/lubridate/issues?q=is%3Aissue+is%3Aopen+mdy

This is a known bug; it looks like they're thinking about doing something about it, but the only workaround is to create an NA: https://github.com/tidyverse/lubridate/issues/685

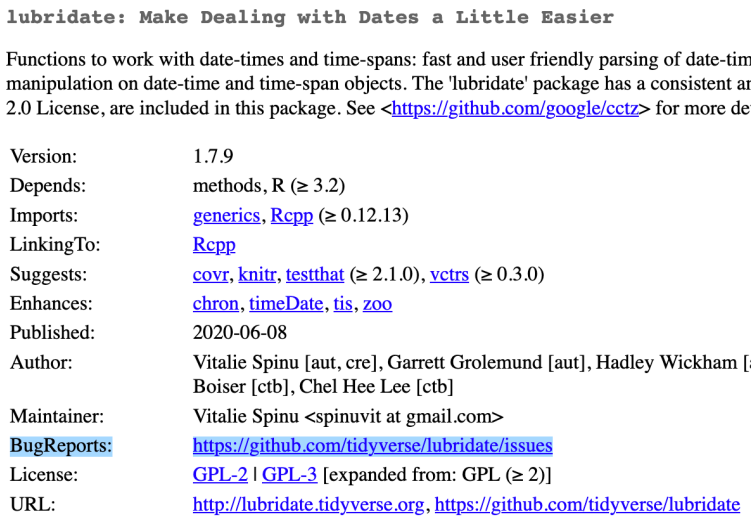
Writing a reproducible example: `reprex`

- https://reprex.tidyverse.org/
- https://reprex.tidyverse.org/articles/articles/learn-reprex.html
- https://reprex.tidyverse.org/articles/reprex-dos-and-donts.html

Do not do these things in your reprex

If you do them, Jenny Bryan will come to your office and set your computer on fire.

- `setwd('/users/danhicks/projects/catsaregreat/myscript/')`



lubridate: Make Dealing with Dates a Little Easier

Functions to work with date-times and time-spans: fast and user friendly parsing of date-tim
manipulation on date-time and time-span objects. The 'lubridate' package has a consistent ar
2.0 License, are included in this package. See <https://github.com/google/cctz> for more de

Version:	1.7.9
Depends:	methods, R (≥ 3.2)
Imports:	generics, Rcpp (≥ 0.12.13)
LinkingTo:	Rcpp
Suggests:	covr, knitr, testthat (≥ 2.1.0), vctrs (≥ 0.3.0)
Enhances:	chron, timeDate, tis, zoo
Published:	2020-06-08
Author:	Vitalie Spinu [aut, cre], Garrett Grolemund [aut], Hadley Wickham [Boiser [ctb], Chel Hee Lee [ctb]
Maintainer:	Vitalie Spinu <spinuvit at gmail.com>
BugReports:	https://github.com/tidyverse/lubridate/issues
License:	GPL-2 | GPL-3 [expanded from: GPL (≥ 2)]
URL:	http://lubridate.tidyverse.org, https://github.com/tidyverse/lubridate

Figure 1: Screenshot of ‘lubridate‘ on CRAN, highlighting the BugReports field

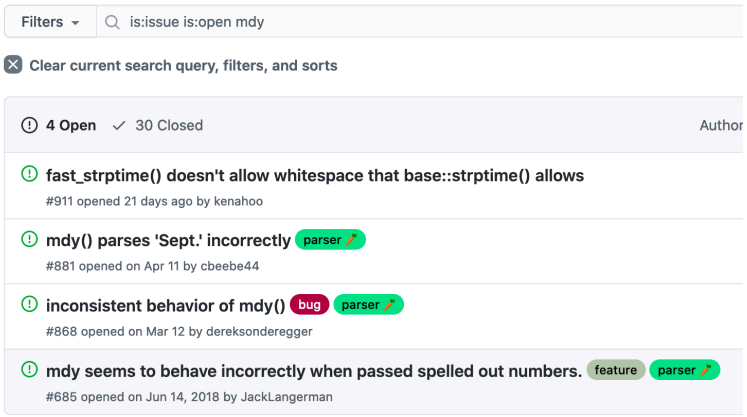


Figure 2: Screenshot of ‘lubridate‘ issues page, showing a relevant search result

    - Used to ensure that R is running where your file is
    - Unnecessary if you're opening different projects in different RStudio sessions
    - Will cause irrelevant errors on any other system
    - Aside: use `file.path()` or `here::here()` to build paths
- `rm(list=ls())`
    - Used because people think it clears out the global environment
    - Unnecessary if you're regularly using Session → Restart R
    - Also unnecessary at the top of a `Rmd` file, which is always knit in a new session
    - Doesn't actually clear out the global environment
        * eg, doesn't unload packages or reset `options()`

Not on Bryan's list, but a bonus thing to not do:

- `require(package)`
    - If `package` is installed, will act just like `library()`
    - If not, will return `FALSE`
        * which generally means the script will keep going until you get some kind of error about a missing function sometime later
        * probably not the error you wanted help with
        * annoying to debug because I have no idea where the function is supposed to come from
    - If `library()` can't find the package, it immediately raises an error
        * I can tell right away what package needs to be installed